



Lecture # 5

Project Management Process Groups and Knowledge Areas

ENGINEERING PROJECT MANAGEMENT

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Value Management

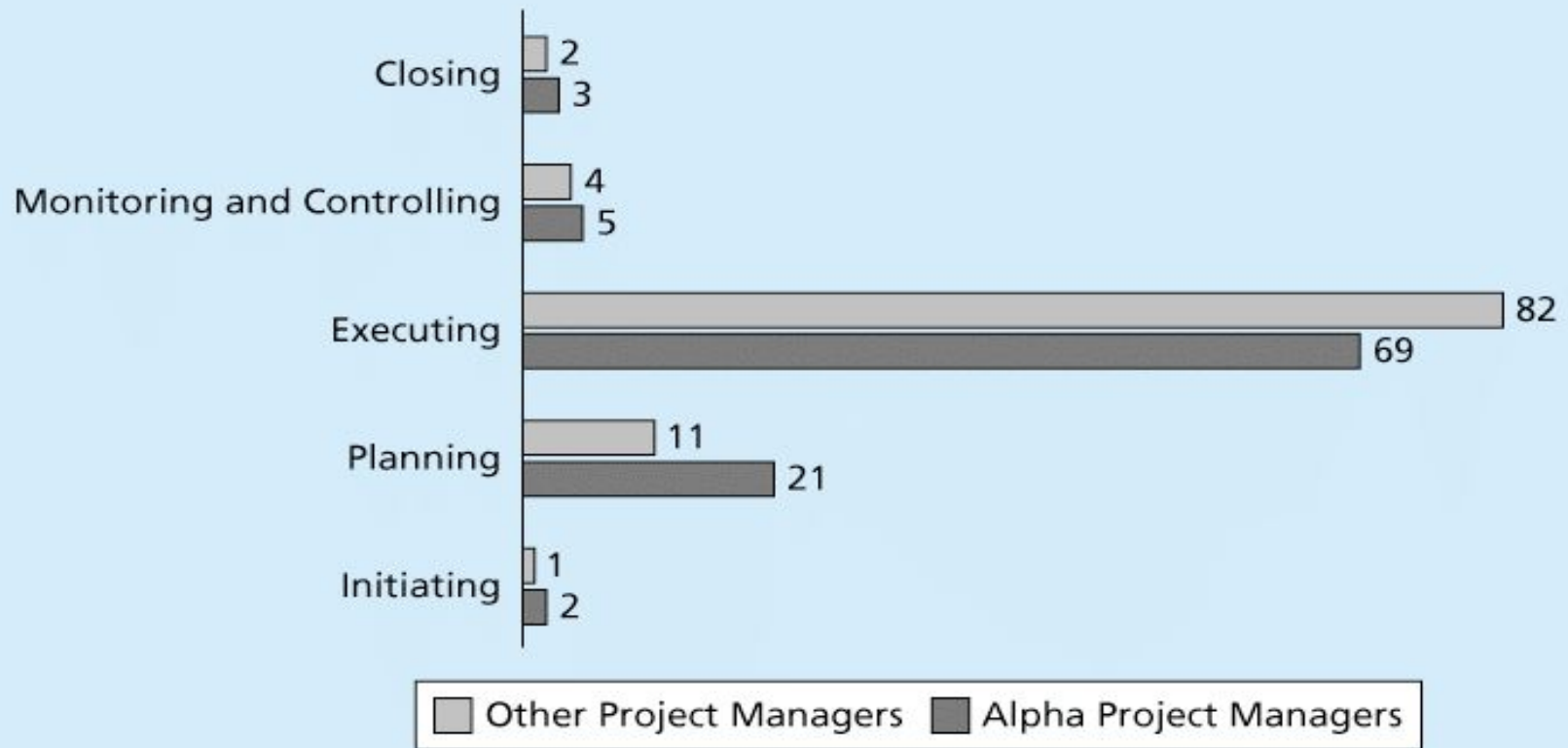
- ▶ A **process** is a series of actions directed toward a particular result
- ▶ Project management can be viewed as a number of interlinked processes
- ▶ The project management process groups include
 - ▶ initiating processes
 - ▶ planning processes
 - ▶ executing processes
 - ▶ monitoring and controlling processes
 - ▶ closing processes

Series of Activities

	WBS ▼	Task Name ▼
1	1	▸ System Development Project
2	1.1	▸ Project Management
3	1.1.1	Planning
4	1.1.2	Cost & schedule management
5	1.1.3	Scope management
6	1.1.4	Task management
7	1.1.5	Project office administration
8	1.1.6	Project communications
9	1.1.7	Human resources management
10	1.1.8	Space & facilities planning
11	1.1.9	Risk management
12	1.1.10	Procurement management
13	1.1.11	Quality management
14	1.2	▸ Systems Engineering
15	1.2.1	Technical planning
16	1.2.2	Technical supervision
17	1.2.3	Business requirements definition
18	1.2.4	System requirements definition
19	1.2.5	System architecture & top level design

20	1.3	▸ Software
21	1.3.1	▸ Software Design
22	1.3.1.1	Software requirements specifications
23	1.3.1.2	Software workpackage definition
24	1.3.1.3	Software prototyping
25	1.3.1.4	Software unit detail design
26	1.3.2	▸ Software Build
27	1.3.2.1	Software unit coding
28	1.3.2.2	Software unit debugging
29	1.3.3	▸ Unit Testing
30	1.3.3.1	Unit test planning
31	1.3.3.2	Unit test case preparation
32	1.3.3.3	Unit test conduct
33	1.3.3.4	Unit test records
34	1.4	▸ Hardware
35	1.4.1	Hardware requirements planning
36	1.4.2	Hardware system definition
37	1.4.3	Hardware component selection
38	1.4.4	Hardware unit/component testing
39	1.5	▸ Deliverables Management
40	1.5.1	Deliverables tracking
41	1.5.2	Deliverables production & packaging

Percentage of time spend on each process group



Suggest that: (1) most time is spent on executing, followed by planning; (2) spending a fair amount of time on planning should **lead** to lesser time spent in execution

PM Process Groups

- ▶ A Project Management Process Group is ---- a **logical_grouping of project management processes** to achieve specific project objectives
- ▶ **Process Groups** are **independent of project phases**. Project management processes are grouped into the following five Project Management Process Groups:

PM Process Groups

- ▶ **1. Initiating Process Group.** Those processes performed to **define a new project** or a new phase of an existing project by **obtaining authorization** to start the project or phase.
- ▶ **2. Planning Process Group.** Those processes required to establish the scope of the project, refine the objectives, and define the course of action required to attain the objectives that the project was undertaken to achieve.

PM Process Groups

- ▶ **3. Executing Process Group.** Those processes **performed** to complete the work defined in the project management plan to satisfy the project requirements.
- ▶ **4. Monitoring and Controlling Process Group.** Those processes required to track, review, and regulate the **progress** and **performance** of the project; identify any areas in which changes to the plan are required; and initiate the corresponding changes.
- ▶ **5. Closing Process Group.** Those processes performed to formally complete or close the project, phase, or contract.

PM Knowledge Areas

- ▶ A Knowledge Area is an identified area of project management defined by its **knowledge requirements** and described in terms of its component processes, practices, inputs, outputs, tools, and techniques.

PM Knowledge Areas

- ▶ The ten Knowledge Areas are:
 - ▶ **1. Project Integration Management.** Includes the processes and activities to identify, define, combine, **unify**, and **coordinate the various processes** and project management **activities** within the Project Management Process Groups.
 - ▶ **2. Project Scope Management.** Includes the processes required to ensure the project includes **all the [project] work required**, and only the work required, to complete the project successfully.

PM Knowledge Areas

- ▶ **3. Project Schedule Management.** Includes the processes required to **manage the time**ly completion of the project.
- ▶ **4. Project Cost Management.** Includes the processes involved in planning, estimating, budgeting, financing, funding, **managing**, and **controlling costs** so the project can be completed within the approved budget.
- ▶ **5. Project Quality Management.** Includes the processes for incorporating the organization's quality policy regarding **planning**, managing, and **controlling** project and product **quality requirements**, in order to meet stakeholders' expectations.
- ▶ **6. Project Resource Management.** Includes the processes to identify, acquire, and manage the **resources needed** for the successful completion of the project.

PM Knowledge Areas

- ▶ **7. Project Communications Management.** Includes the processes required to ensure timely and appropriate planning, **collection**, creation, distribution, **storage**, **retrieval**, management, control, monitoring, and **ultimate disposition of project information** .
- ▶ **8. Project Risk Management.** Includes the processes of conducting risk management planning, identification, analysis, response planning, response implementation, and monitoring **risk on a project**.
- ▶ **9. Project Procurement Management.** Includes the processes necessary to **purchase or acquire** products, services, or results needed from outside the project team.

PM Knowledge Areas

- ▶ **10. Project Stakeholder Management.** Includes the processes required to identify the people, groups, or organizations that could impact or be impacted by the project, to **analyze stakeholder expectations** and their impact on the project, and to develop appropriate management strategies for **effectively engaging stakeholders** in project decisions and execution.

Discussion

